CLINICAL · EAR

Ototoxicity, noise, presbycusis and immune inner ear disease

Most slowly progressive sensorineural loss in both ears comes from age, noise or drugs. Ask about each specifically, stop whatever is still doing harm, and do not miss the rarer autoimmune inner ear disease, because it can respond to steroids.

AT A GLANCE

- Think age, noise and drugs.
- Stop harmful drugs and noise where you can.
- Noise gives a 4 kHz notch.
- Rapid bilateral loss suggests immune disease: try steroids.
- UK action values 80 and 85 dB(A) with a 3 dB exchange rate; CLB criteria for noise loss; synaptopathy still debated.

1 CHAPTER ONE Overview

Four causes of acquired sensorineural loss.

1.1 Scope

- **Ototoxicity:** hearing or balance damaged by drugs.
- **Noise-induced hearing loss:** from sustained or sudden loud sound.
- **Presbycusis:** age-related hearing loss.
- **Immune-mediated inner ear disease:** primary or as part of systemic disease.

1.2 Why it matters

Much of this hearing loss is preventable, by monitoring drugs and protecting ears from noise, and some is reversible.

2 CHAPTER TWO Ototoxicity

Drugs that damage the ear.

2.1 The drugs

OTOTOXIC DRUGS

DRUG	MAIN TARGET	REVERSIBLE?
Streptomycin, gentamicin	Vestibular	Often permanent
Neomycin, kanamycin, amikacin	Cochlea	Often permanent
Loop diuretics	Stria vascularis	Usually
Salicylates	Cochlea	Yes
Quinine, chloroquine	Cochlear vessels	Usually; high doses may not be
Cisplatin, carboplatin	Outer hair cells	Often permanent

Others include deferoxamine, some ear drops (aminoglycosides and polymyxin) through a perforation, and some topical antiseptics.

WHERE THE BOOKS DIFFER · TOBRAMYCIN

Dhingra
Mainly vestibulotoxic.

Lee
Mainly cochleotoxic.

2.2 Aminoglycosides

Hair cells are damaged from the basal turn upwards, so high frequencies go first. Hearing loss may appear during or after treatment. Risk is higher with renal impairment, age over 65, high blood levels, long courses, previous exposure, loop diuretics, and genetic susceptibility: mitochondrial 12S rRNA mutations make even a single dose dangerous, so ask about family members who went deaf after these drugs.

2.3 Cisplatin

Bilateral, permanent high-frequency loss, worse in young children under five, the elderly, and with noise or radiotherapy.

2.4 Prevention

Avoid ototoxic drugs where alternatives exist, monitor levels and renal function, and obtain baseline and serial audiograms, including high-frequency audiometry.

ON THE WARD

New tinnitus, hearing loss or imbalance on an aminoglycoside or cisplatin: alert the treating team, check levels and renal function, and test hearing.

2.5 Deliberate use

Gentamicin's vestibular toxicity is used therapeutically in Ménière's disease.

3 CHAPTER THREE Noise-induced hearing loss

Loud sound over time.

3.1 Mechanism

Outer hair cells in the basal turn fail first, around the region that responds to 4 kHz.

3.1.1 Temporary threshold shift

Hearing dulls after loud exposure and recovers.

WHERE THE BOOKS DIFFER · RECOVERY FROM A TEMPORARY SHIFT

Dhingra	Lee
Up to two weeks.	Within 24 hours.

3.1.2 Permanent threshold shift

Repeated exposure leads to irreversible loss.

3.1.3 Acoustic trauma

A single blast or explosion (around 140 dB or more) can rupture the drum, disrupt the ossicles and damage the cochlea at once.

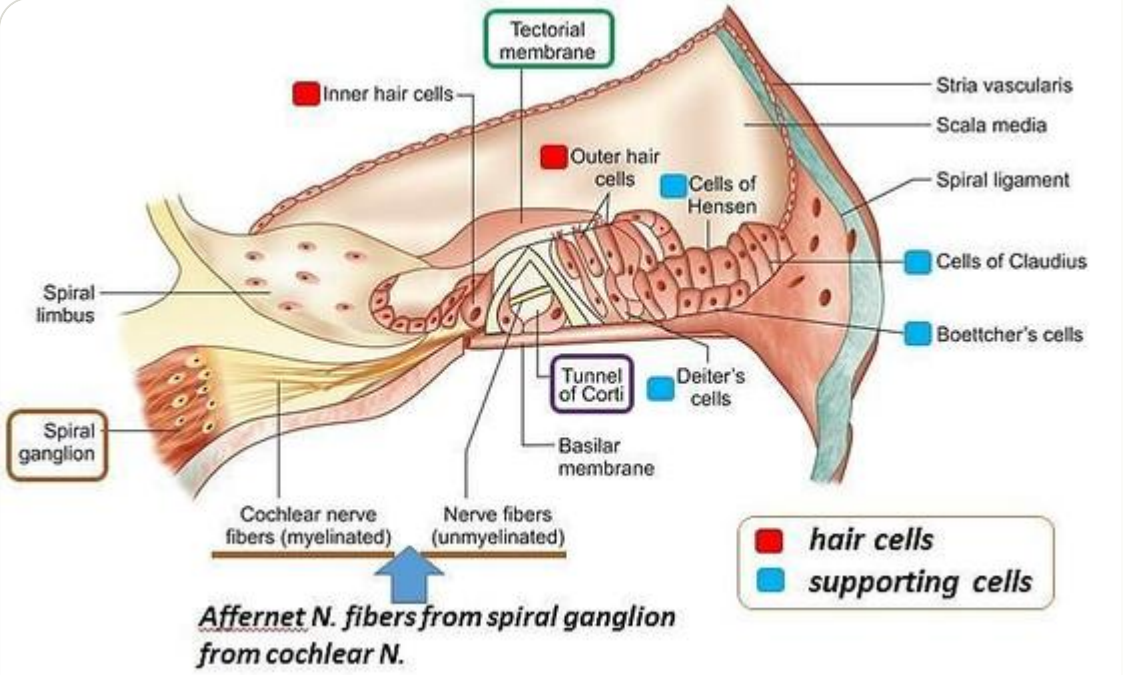


FIGURE 1 The organ of Corti: outer hair cells are the first casualties of noise and ototoxic drugs.

Anatomy atlas · ENT Steps p. 77

3.2 Clinical picture

High-pitched tinnitus and difficulty hearing in noise. The audiogram shows a symmetrical *notch at 4 kHz*, which deepens and widens into the speech frequencies with continued exposure.

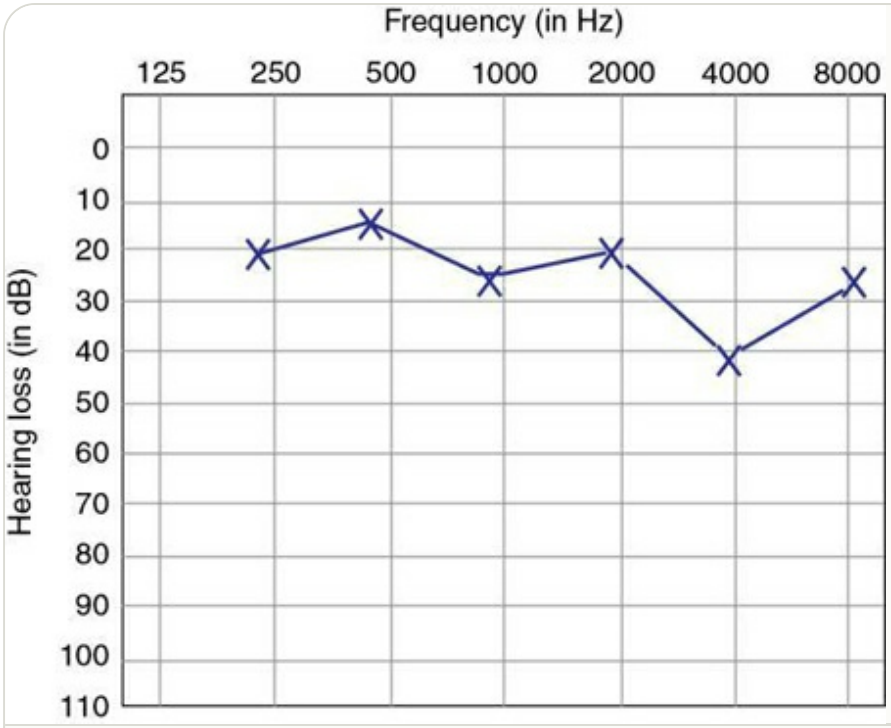


FIGURE 2 Early noise-induced hearing loss, with a dip at 4000 Hz.
Dhingra, fig. 5.7

3.3 Safe limits

NOISE EXPOSURE RULES

- Dhingra's occupational limit: 90 dBA for 8 hours; no continuous exposure over 115 dBA; no impulse over 140 dBA.
- The *5 dB rule*: each 5 dB rise halves the safe duration.
- Hearing conservation (protection and audiometry) starts at 85 dBA.
- Limits differ by country; check local regulations.

3.4 Prevention and management

Remove from exposure, reduce noise at source, and use hearing protection: earplugs give about 15 to 30 dB, earmuffs 30 to 40 dB, and both together more. Protectors work better at high than low frequencies. Pre-employment and annual audiograms. Hearing aids for established loss.

WHERE THE BOOKS DIFFER · EXPOSURE LIMITS AND THE EXCHANGE RATE

Dhingra

Occupational limit of 90 dBA for 8 hours, with each 5 dB rise halving the safe duration.

Oxford

The UK Control of Noise at Work Regulations 2005 set a lower action value of 80 dB(A), when protection must be offered and information given, and an upper action value of 85 dB(A), when protection is compulsory and hearing surveillance required, with an absolute limit of 87 dB(A) at the ear allowing for protection. European and UK practice uses a 3 dB exchange rate (each 3 dB doubles the energy and halves the safe time); the 5 dB rule is the older American occupational convention.

DIAGNOSIS, ACOUSTIC TRAUMA AND HIDDEN HEARING LOSS

Noise first stiffens and then kills the **outer hair cells**, producing a temporary threshold shift that becomes permanent with continued exposure; 3, 4 and 6 kHz are affected first. In medicolegal practice the **Coles, Lutman and Buffin (CLB) guidelines** require high-frequency loss beyond that expected for age, a cumulative noise immission level of at least 100 dB (roughly 85 dB(A) daily for many years), and a notch or bulge at 3 to 6 kHz at least 10 dB deeper than the thresholds at 1 or 2 kHz and at 6 or 8 kHz. **Acute acoustic trauma** from a single blast or impulse causes immediate loss without a preceding temporary shift and is treated like sudden sensorineural loss, with steroids. Animal work by Kujawa and Liberman showed that noise can destroy the synapses between inner hair cells and the cochlear nerve while thresholds recover (**cochlear synaptopathy**, or "hidden hearing loss"), which may explain difficulty in noise with a normal audiogram; in humans the effect appears variable and is still debated. Recreational noise, from music players, concerts and shooting, is an increasingly important cause in young people.

4 CHAPTER FOUR Presbycusis

Hearing and age.

4.1 Picture

Gradual, bilateral, symmetrical high-frequency loss. The patient hears speech but cannot understand it, especially in noise, and is intolerant of loud sound because of recruitment.

4.2 Types (Schuknecht)

TYPES OF PRESBYCUSIS

TYPE	DAMAGE	AUDIOGRAM	DISCRIMINATION
Sensory	Hair cells, basal turn	Steep high-tone loss	Good
Neural	Cochlear neurones	Variable	Poor, out of proportion
Strial (metabolic)	Stria vascularis	Flat	Good
Cochlear conductive	Stiff basilar membrane	Gently sloping	Moderate

4.3 Management

Hearing aids, assistive listening devices, speech reading and communication tactics; cochlear implants for severe loss. Untreated hearing loss is linked with isolation and cognitive decline.

OPERATIVE GUIDES IN THEATRE 2 operations

Cochlear Implantation in Adults
Cochlear implant for severe or profound loss

Implantable Middle Ear Devices
Implantable devices

CHAPTER FIVE

5 Immune-mediated inner ear disease

The treatable one.

5.1 Primary autoimmune inner ear disease

Rapidly progressive, bilateral, often asymmetric sensorineural loss over weeks to months, usually in middle age (40 to 50), with vestibular symptoms in about half. About 15% have another autoimmune disorder.

5.2 Secondary

Part of systemic disease: *Cogan syndrome* (interstitial keratitis with audiovestibular loss), granulomatosis with polyangiitis, systemic lupus, antiphospholipid syndrome, and others.

5.3 Diagnosis

Bilateral loss of 30 dB or more with progression on serial audiograms within three months. Blood tests: ESR, rheumatoid factor, ANA, complement; anti-Hsp70 antibodies correlate with activity but are not specific. Contrast MRI to exclude other causes. A response to steroids supports the diagnosis.

5.4 Treatment

Prednisolone 1 mg/kg/day, up to 60 mg, for four weeks. Responders (an improvement of 15 dB at one frequency or 10 dB at consecutive frequencies, or better discrimination) continue to a plateau and taper to the lowest maintenance dose over months. For steroid intolerance or dependence: methotrexate, cyclophosphamide (more toxic), biological agents or intratympanic steroids. Cochlear implantation if hearing is lost.

6 CHAPTER SIX Differential diagnosis

Others to exclude.

- Vestibular schwannoma.
- Ménière's disease.
- Syphilis.
- Multiple sclerosis.
- Familial progressive sensorineural loss.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Ototoxic drugs:** vestibulotoxic versus cochleotoxic, risk factors, monitoring.
- **Noise:** 4 kHz notch, temporary and permanent shifts, safe limits and the 5 dB rule.
- **Presbycusis:** Schuknecht's four types.
- **Autoimmune inner ear disease:** picture and steroid trial.

7.2 Common pitfalls

COMMON PITFALLS

- Not asking about drugs and noise.
- Missing a mitochondrial susceptibility to aminoglycosides in the family history.
- Calling rapidly progressive bilateral loss presbycusis.

Written for this app from Oxford Textbook of Otolaryngology ch. 11, 17, Oxford Specialist Handbook ch. 18, Oxford Handbook of ENT ch. 7, Dhingra ch. 5 and K. J. Lee ch. 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

INNER EAR & IAC · BONE DISEASE

Labyrinthitis Ossificans (Ossification after Meningitis)

IMAGING FINDINGS

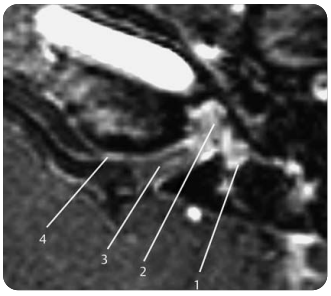
- Loss of the bright fluid signal of the cochlea on high-resolution T2
- Later: bone in the basal turn and canals on CT
- Early: enhancement of the labyrinth

TOP DIFFERENTIALS

- Labyrinthine or cochlear aplasia
- Cochlear otosclerosis
- Intralabyrinthine schwannoma

CHECKLIST

- After meningitis: urgent MRI and cochlear implantation
- Report patency of the basal turn



MRI, T1-weighted with gadolinium enhancement, axial. There is enhancement of the vestibule (1), cochlear contents (2), and the…



Axial bone CT in a 2-year-old boy with bilateral sensorineural hearing loss

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116.